SnakeGuard

Problem Statement, Proposed System, and Literature Review

1. Problem Statement

Snakebite is one of the most under-addressed occupational hazards in agriculture. The World Health Organization estimates that roughly five million people are bitten by snakes every year worldwide, with 1.8 to 2.7 million of those bites resulting in envenoming and up to 138,000 deaths annually. The overwhelming majority of these incidents occur in rural farming communities, where people spend entire working days at ground level, in tall crops, dense grass, or low dawn-and-dusk light — precisely the conditions in which a snake's camouflage is most effective and a human eye is least reliable.

The people most exposed to this risk are also the people least protected against it. Protective footwear and awareness training reduce risk, but neither solves the underlying problem: a snake that a farmer cannot see is a snake the farmer cannot avoid. Protective clothing is often impractical in hot climates and, even where worn, only mitigates some bites rather than preventing close encounters altogether. Snakes are cryptic by design — their camouflage defeats casual visual detection in crop rows and undergrowth, especially during the low-light hours when both snakes and farmers are most active.

No affordable, real-time, wearable detection device exists in common agricultural use today. Where camera-based safety systems do exist in adjacent domains — farm perimeter security, roadside wildlife monitoring — they are typically opaque: an alarm fires, or fails to fire, with no way to verify why. A warning device nobody can trust is a warning device nobody keeps wearing.

A closer look at the existing snake-detection and identification literature (Section 3) reveals a second, more technical problem underlying this gap: the datasets used to train these systems are almost always composed exclusively of snake images. Of the eight prior works reviewed for this project, only one trains its detector against snake-like negative objects — ropes, sticks, and leaves — and even that system stops short of confirming detections across multiple frames or scoring danger on a graded scale. The remaining seven either omit look-alike negatives altogether or do not disclose their training composition. A model trained only on “snake” and “not snake” as abstract classes, without ever seeing the specific objects it is most likely to confuse with a snake, will alarm on a garden hose as readily as on a viper. This is precisely the failure mode SnakeGuard is designed to close: our training data deliberately includes ropes, tree roots, sticks, and other visually similar objects as explicit negative and confusable classes, rather than treating “non-snake” as an unstructured background category.

2. Proposed System

SnakeGuard is an ankle-worn, camera-based early-warning system that watches the ground immediately ahead of a farmer's next step, distinguishes a real snake from common look-alike objects, judges how

dangerous it is, and turns that judgement into a sound simple enough to react to without stopping, looking down, or reading a screen — a beep that quickens as danger increases.

The system is built as a five-stage pipeline that runs continuously while the device is worn.

Stage 1 — Sense

An ankle-mounted camera continuously observes the ground-level area directly around the wearer's feet, capturing the same low, cluttered, low-light field of view in which snake encounters actually happen — rather than a chest- or head-mounted view that misses ground-level threats entirely.

Stage 2 — Detect

A trained object-detection model separates a real snake from common field look-alikes — ropes, coiled hoses, sticks, and tree roots — in every incoming frame. This is the safety-critical foundation of the whole system: every downstream decision (venom classification, proximity estimate, alert level) depends on this stage being right, so it is trained and evaluated with a dataset that explicitly includes these confusable objects as their own classes rather than lumping them into an undifferentiated background.

Stage 3 — Classify

Once a detection is confirmed as a snake, a second model examines it to judge venomous versus non-venomous status, so that the urgency of the alert reflects the actual level of danger rather than treating every snake sighting identically.

Stage 4 — Locate

The size of the detected snake within the frame — its bounding-box area — is used as a proxy for distance: a larger bounding box means the snake is closer underfoot, allowing the system to estimate proximity without additional depth-sensing hardware.

Stage 5 — Alert

An onboard buzzer converts the detection, classification, and proximity estimate into a single, glanceless signal: the beep rate increases as the snake gets closer, and an immediate, distinct alert is issued if the snake is judged venomous — giving the wearer an instant warning without requiring them to interpret a screen or a written message.

Explainability and Validation Layer

Before any alert can be trusted, the detector itself has to be. The Detect stage is the safety-critical foundation every downstream decision depends on — classification, proximity estimation, and alerting are all meaningless if the underlying judgement of “is this a snake” cannot be trusted. Standard deep object detectors are black boxes: they output a confidence score with no account of why that score was reached, and for a device that decides whether to alarm a farmer in a field, a single number is not sufficient grounds for trust. It cannot, on its own, catch a spurious correlation, confirm genuine reasoning, or explain a failure

mode before the device is deployed. For this reason, the project does not rely on any one explanation technique or on a single confidence score; instead it combines several independent methods designed to cross-check one another, so that the blind spots of one are covered by the strengths of another, with a quantitative test used to settle any disagreement between them objectively rather than by visual judgement alone.

The plan combines activation-based and gradient-based visual explanation methods as a first layer of cross-checking. An activation-based method (Eigen-CAM), which needs no gradients or class target, is used as a fast, always-available first-pass check of which image regions the detector is generally responding to. This is paired with a gradient-based, class-targeted method (Grad-CAM++), which can answer the more specific question of why a given detection was labelled a snake rather than merely what looks salient in the frame; agreement between an activation-only method and a gradient-based method is treated as stronger evidence than either alone. Because YOLO-family detectors read from multiple feature-map scales simultaneously to detect objects of different sizes, the gradient-based explanation is planned to fuse information across scales rather than hook only the coarsest layer nearest the output — a single-scale hook risks missing the fine spatial detail that actually distinguishes a thin, elongated object such as a snake from its surroundings.

Since both of the above methods still look inside the network and therefore share some of its assumptions, an independent, model-agnostic method — occlusion sensitivity — is included as a ground-truth cross-check: it systematically hides small patches of the input and measures how much the model's confidence drops when each patch is hidden, without any assumption about architecture, gradients, or which layer to inspect. Finally, rather than judging any heatmap by eye, a deletion/insertion AUC test is planned to turn “this explanation looks plausible” into an objective score — pixels are progressively removed or revealed in the order each method claims is most important, and the resulting confidence curve is compared against a random-order baseline. A trustworthy explanation should collapse confidence quickly on deletion and recover it quickly on insertion, and this quantitative step is what is intended to decide which explanation method is actually reliable, rather than assuming the best-looking heatmap is the correct one.

Because the training data already includes a confusable negative class — ropes, roots, and other look-alike objects — the explainability layer is designed to be tested specifically on these hardest cases, comparing matched pairs of real snakes and look-alikes side by side. This is intended to surface concrete, human-checkable reasons behind both correct detections and confusions — for instance, whether a real snake is correctly drawing attention to its coiled body or head shape, and whether a look-alike that fools the model does so because it happens to share a locally snake-like texture, such as a cluster of thin, dark, curved strands. Framed this way, a misclassification becomes a legible, diagnosable error rather than an unexplained black-box failure. Standard attribution techniques such as SHAP and LIME were considered but set aside for this stage, since both were built around single-label image classification and are a poor structural fit for object detection, where one frame can contain multiple boxes, classes, and confidence scores; the methods above were chosen instead because they are built for, or have been shown to work on, detection models specifically.

This explainability layer directly targets the opacity problem identified in the Problem Statement: a wearer can only trust an alert if the system's reasoning has itself been examined and validated, not merely if its output accuracy looks acceptable on a held-out test set.

Taken together, these five stages plus the explainability layer form a pipeline that goes beyond single-purpose detection: it filters out the everyday objects that make existing systems unreliable in the field, grades the danger of a genuine detection rather than treating all snakes alike, estimates how close the threat is, and delivers that judgement through a channel a working farmer can act on without breaking stride.

3. Literature Review

The matrix on the following page summarizes eight prior works relevant to this project: five peer-reviewed papers (References 1–5) and three patent-style prior-art disclosures (References 6–8). The patent disclosures reveal less technical detail than the published papers, since they are filed as intellectual-property documents rather than research papers, but they are treated as part of the same literature base for the purposes of gap analysis below.

Comparative Synthesis

Across the eight works reviewed, a clear methodological trend emerges: single-stage object detectors — almost exclusively members of the YOLO family (YOLOv5, YOLOv8, YOLOv8n, YOLOv8-nano, and unspecified YOLO variants in References 6 and 7) — dominate the field, occasionally supplemented by a separate CNN classifier such as DenseNet121 for venom status (Reference 2). Detection accuracy has improved steadily as datasets have grown, from 71% accuracy on 1,500 images (Reference 3) to a 90.2% mAP on 25,000 images (Reference 4), a 93.4% mAP on an embedded ESP32-CAM platform (Reference 8), and a 94.5% mAP on a curated 12,000-image, 35-species dataset (Reference 5). This improvement has been driven almost entirely by scaling the positive class — more snake images, more species, more environmental variety — rather than by addressing what the snake actually needs to be distinguished from in a real field of view.

The alert-mechanism side of the literature has diversified more than the detection side: simple messaging alerts (WhatsApp in Reference 1), LED/buzzer/SMS alerts (Reference 6), MQTT-based buzzer alerts (Reference 8), proximity-based warning triggers (Reference 7), and GPS-geotagged cloud dashboards with emergency-service coordination (References 4 and 5). What almost none of these eight works do is treat snake-like objects as a first-class part of the training problem itself. Reference 8 is the sole exception, explicitly training on ropes, sticks, and leaves alongside snakes — yet even it stops at a single confidence threshold, without the temporal confirmation, graded threat scoring, or explainability validation that would make such a detector trustworthy for a wearable, hands-free device. Whether a system reports 71% accuracy or 94.5% mAP, the figure describes performance on curated positive and generic negative examples, not on the confusable field clutter that actually triggers false alarms in agricultural settings. This is a dataset-design and validation gap, not simply a capacity or architecture gap: a larger or more accurate model trained the same way will still fail in the same way.

Research Gap

- Dataset gap: Seven of the eight reviewed works (References 1–7) train on snake-only positive datasets without a curated, explicit negative class of visually similar field objects (ropes, hoses, sticks, tree roots). Only Reference 8 includes such negatives, and even it lacks the downstream stages described below.
- Methodological gap: None of the eight works — including Reference 8 — combine single-frame detection with multi-frame temporal confirmation before triggering an alert, leaving all of them vulnerable to one-off false positives from a single ambiguous frame.

- Threat-scoring gap: Existing systems report a detection confidence, a proximity trigger (Reference 7), or at best a binary venomous/non-venomous label (References 2, 5); none compute a graded, continuous threat score that combines venom status with estimated proximity into a single actionable signal.
- Explainability gap: None of the eight reviewed systems validate what the model is actually attending to (e.g., coiled shape, head silhouette) versus incidental background cues, leaving their false-positive and false-negative behavior largely unexplained.
- Application/hardware gap: The reviewed systems are either handheld mobile apps (References 3, 5), fixed/edge installations (References 1, 4, 6, 7), or a fixed embedded sensor node (Reference 8); none is a hands-free, ankle-worn wearable that operates continuously while the user's attention and hands remain free for agricultural work.

4. Identifying Common Themes

Grouping the eight reviewed works by their core technical approach, rather than by publication date, makes the shape of the existing literature — and the gap SnakeGuard addresses — easier to see.

Group A: Fixed or Edge-Camera Detection with Automated Alerting

- Reference 1 — Shivaraj et al., Real-Time Snake Detection and Alert System (YOLOv8 + WhatsApp)
- Reference 4 — Limje et al., IoT-Enabled Continuous Snake Detection System (YOLOv8n + GPS dashboard)
- Reference 6 — IN 202521121126, Real-Time Animal Threat Identification and Alert System
- Reference 7 — CN 121459270, Forest Monitoring System Based on Internet of Things

Group B: Mobile or Web-Based Species and Venom Identification

- Reference 2 — Prasad et al., VenoScan (DenseNet121 web app)
- Reference 3 — Dube & Bhuru, Snake Identification System Using CNNs (YOLOv5 Android app)
- Reference 5 — Bharath et al., Serpent-ID (YOLOv8-nano Flutter app with first-aid and emergency modules)

Group C: Embedded Sensor Nodes with Explicit Negative-Object Training

- Reference 8 — IN 202541120085, Low-Latency Snake Detection Using Embedded CNN and MQTT-Based IoT Alerts

This grouping becomes the structure of the Research Gap above: Group A shows that alerting infrastructure is mature but detection remains single-class and untested against look-alike clutter; Group B shows that identification accuracy and user-facing features (first aid, emergency contact) are well developed but confined to handheld, non-continuous use; and Group C, though it stands alone, demonstrates that negative-object training is technically straightforward to add — it has simply not yet been combined with temporal confirmation, graded threat scoring, or a wearable, hands-free form factor. SnakeGuard is positioned directly at the intersection of these three groups, taking the alerting maturity of Group A, the identification depth of

Group B, and the negative-class training strategy of Group C, and combining them with the temporal-confirmation, threat-scoring, and explainability layers that none of the eight reviewed works implement.

Literature Review Matrix

No.	Reference	Problem Addressed	Dataset	Methodology	Results	Limitations	Key Contribution
1	Shivaraj et al., "Real-Time Snake Detection and Alert System," IJERT, Vol. 13 Issue 5, 2024	Real-time snake detection in video and automated alerting to reduce human–snake conflict	~1,000 CVAT-annotated snake images, converted to YOLO format	YOLOv8 object detection on live video; WhatsApp alerts via Pywhatkit	All-class F1 ≈ 0.65 (peak at conf. 0.254); precision reaches 1.00 at conf. 0.844; distinguishes venomous vs non-venomous	Dataset contains only snake images — no rope/stick/root negatives, so false-positive behaviour is untested; single-frame detection with no temporal confirmation; no distance or threat scoring	Real-time YOLOv8 pipeline integrated with automated WhatsApp alerting
2	Prasad et al., "VenoScan: AI-Powered Classification of Venomous and Non-Venomous Snakes," ISACC, 2025	Venom-status classification of snake images to assist doctors/forest department with treatment decisions	Kaggle "Snake Dataset-India," 2,043 images (cobras, vipers, rat snakes, green tree vine)	Fine-tuned DenseNet121 (GAP → Dense-256-ReLU → Dropout 0.5 → Dense-1-Sigmoid); deployed as a Visual Studio web app	Reported test accuracy 98.73% (abstract) vs 82.73–90% (results section); precision 0.92/0.87, recall 0.86/0.93, F1 0.89/0.90	Static-image classification only — no real-time detection, no environmental monitoring, no snake-vs-object filtering, no wearable/hardware deployment, no multi-stage pipeline; limited to 4 snake types	Image-based venom classifier deployed as an accessible web tool for non-experts
3	Dube & Bhuru, "Snake Identification System Using Convolutional Neural Networks," ZCICT, 2022	Public misidentification of snake species leading to mishandling/envenoming; need for real-time mobile identification	Custom dataset, 1,500 images, 8 Southern African species, annotated in MakeSense.ai	YOLOv5 → converted to TensorFlow Lite → deployed in an Android (Java) mobile app	71% overall accuracy; mAP 0.713@0.5; all-class F1 ≈ 0.71; frequent confusion between visually similar species (Mole Snake vs Night Adder)	Limited classification accuracy; small dataset (~150 images/class); no false-positive/look-alike object handling; no threat assessment, wearable IoT, or adaptive alert hardware	First end-to-end pipeline from a trained CNN to a real-time on-device mobile identification app
4	Limje, Reddy & Sivakumar B, "An IoT-Enabled Continuous Snake Detection System with GPS-Based Alerting for Wildlife Conservation," ICCSP, 2026	Lack of lightweight, deployable, real-time edge systems for snake detection with alerting and geolocation	~25,000 annotated images from multiple public/curated sources (incl. Kaggle Snake Species Dataset), diverse habitats and lighting	YOLOv8n on a Raspberry Pi 5 edge device; GPS geo-tagging; Node/Firebase backend; Next.js monitoring dashboard	mAP@0.5 = 90.2%; F1 = 84%; 15–20 FPS on-device inference	Single-class "snake" detection only — no venom classification, no explicit filtering of snake-like objects; performance affected by lighting, motion blur, and occlusion; GPS/battery constraints in remote areas	Edge-deployable real-time detection with GPS geo-tagging and a cloud dashboard for conservation authorities
5	Bharath et al., "Serpent-ID: A Deep Learning-Based Mobile Application for Rapid Snake Identification and Bite Management Guidance," ICSFT, 2026	Rapid species identification plus WHO-compliant first aid and emergency coordination for snakebite victims	12,000 images, 35 medically significant Indian species (SnakeCLEF 2023 + public repositories)	Quantized YOLOv8-nano (INT8, TFLite) in a Flutter mobile app; GPS + Google Maps hospital locator; first-aid module	mAP@0.5 = 94.5%, 28 ms inference, 5.8 MB model; validation (n=200): 94% species-ID accuracy, 100% correct first-aid display, 98% emergency-link success	Covers only 35 species; accuracy drops with <100 training images/species or poor conditions; hospital database needs manual updates; no dedicated hard-negative class for snake-like objects; no	Fully integrated mobile pipeline combining detection, first aid, and emergency-service coordination

No.	Reference	Problem Addressed	Dataset	Methodology	Results	Limitations	Key Contribution
						independent venom-classification stage	
6	IN 202521121126, “A Real-Time Animal Threat Identification and Alert System for Agricultural Fields” (filed 2026-01-16)	General farm-animal threat identification and alerting	Not disclosed	Fixed-installation camera + Raspberry Pi + YOLO-based edge inference; dangerous/non-dangerous threat-assessment module	Not disclosed	General animal (not snake-specific) detection; no snake-vs-look-alike filtering, small-object filtering, temporal confirmation, graded threat score, or venom classification; not wearable	Fixed farm-security system with LED/buzzer/SMS alerts based on animal threat classification
7	CN 121459270, “A Forest Monitoring System and Method Based on Internet of Things” (filed 2026-02-03)	Roadside/forest snake monitoring and pedestrian warning	Not disclosed	IoT roadside camera and audio acquisition; snake recognition, distance-determination, and warning-trigger modules	Not disclosed	Fixed roadside installation, not wearable; no snake-vs-object conflict resolution, small-object filtering, sliding-window temporal confirmation, graded threat score, or venom classification	Proximity-based snake warning with a pedestrian-detection interference mechanism
8	IN 202541120085, “A Low-Latency Snake Detection and Warning System Using Embedded CNN and MQTT-Based IoT Alerts” (filed 2025-12-26)	Continuous, low-cost embedded environmental snake monitoring	Dataset explicitly includes snakes and non-snake look-alikes (ropes, sticks, leaves)	ESP32-CAM + YOLOv5, confidence-threshold scoring, MQTT protocol alerts, buzzer	93.4% mAP	Single confidence threshold only, no bounding-box overlap/conflict resolution; no small-object area filtering; no 3-of-5 sliding-window temporal confirmation; no graded 0–100 threat score; no venom classification or dual-LED indicators; fixed camera, not wearable	Closest prior art to a negative-class strategy, but lacks temporal confirmation and graded scoring